

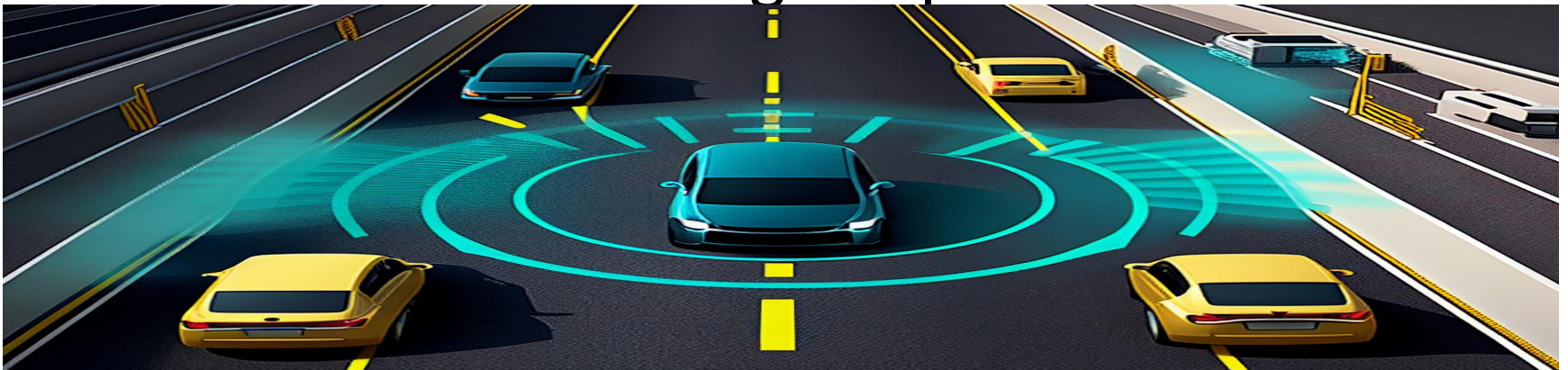
Chapter One

Introduction to Artificial Intelligence



Learning Objectives

- Understand AI definition, history, evolution
- Differentiate AI types
- Identify real-world AI applications
- Learn about AI agents and PEAS model
- Overview of Knowledge Representation



Introduction to Artificial Intelligence (AI)

- Is the branch of computer science.
- Enables machines to perform tasks that normally require human intelligence .
- These include
 - ❖ Problem-solving
 - ❖ Learning
 - ❖ Reasoning,
 - ❖ Perception, and language understanding.

Goals of AI

- Simulate human intelligence
- Create systems that can perform tasks autonomously
- Learn and adapt from experience
- Solve complex problems in real time

AI vs automation

- Why AI not automation?
 - Learning-based: AI systems learn from data and experience.
 - Adaptability: Can handle new, unexpected situations.
 - Decision-making: AI can make predictions or take actions without explicit programming for every case.
- Examples:
 - Self-driving car adapting to traffic conditions
 - ChatGPT answering new questions it has never seen before
 - Google Translate learning from millions of examples

Automation

- Rule-based: Follows fixed, pre-programmed instructions.
- No learning: It cannot adapt or improve unless a human changes the rules.
- Examples:
 - Washing machine cycles
 - ATM cash withdrawal process
 - A script that copies files every night

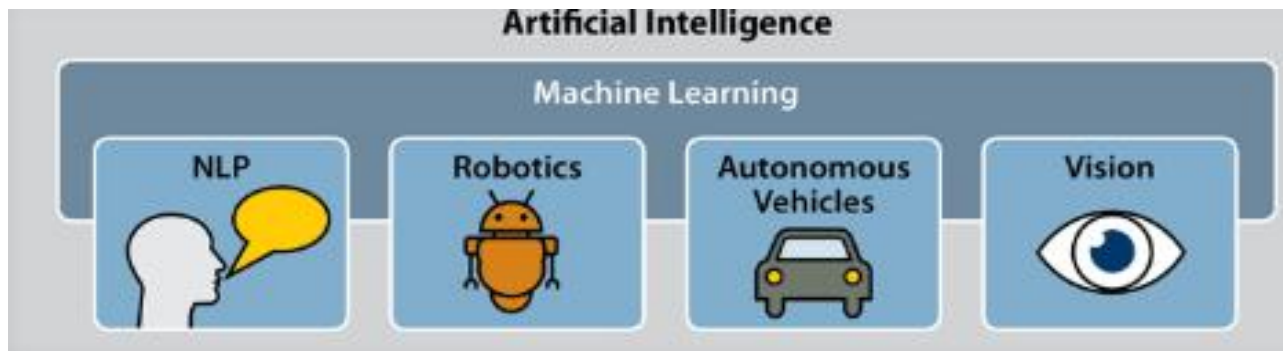
Note:- Automation is rule-based, while AI involves learning and adapting from data.

Automation vs AI

- Automation = Doing repetitive tasks faster (no thinking).
- AI = Thinking + Learning + Adapting like humans.
- Automation = A calculator → it only works with rules you give.
- AI = A math tutor → it learns, adapts, and explains solutions in different ways.

Examples of AI?

- Voice assistants (Siri, Alexa)
- Face recognition systems
- Autonomous vehicles
- Language translation apps
- Medical diagnosis systems



Approaches to AI

- Approaches to AI can be categorized based on how machines are designed to mimic human cognition and behavior.
- The main approaches include
 - thinking like a human
 - acting like a human
 - thinking rationally
 - acting rationally



Approaches to AI

- **Thinking Humanly:** Focuses on modeling how humans think, like with cognitive science. Example: Simulating memory or problem-solving like a human would.
- **Acting Humanly:** Focus is on behaving like a human — even if it doesn't think like one. Measured by passing the Turing Test.
- **Thinking Rationally:** Uses formal logic to make perfect decisions. Old-school expert systems, but not very flexible.
- **Acting Rationally:** The most modern and widely used. Focuses on choosing the best possible action for achieving goals. Based on the rational agent model (like the diagrams you've shared earlier). combines adaptability and learning, making it powerful but raising ethical concerns about decisions that lack empathy.

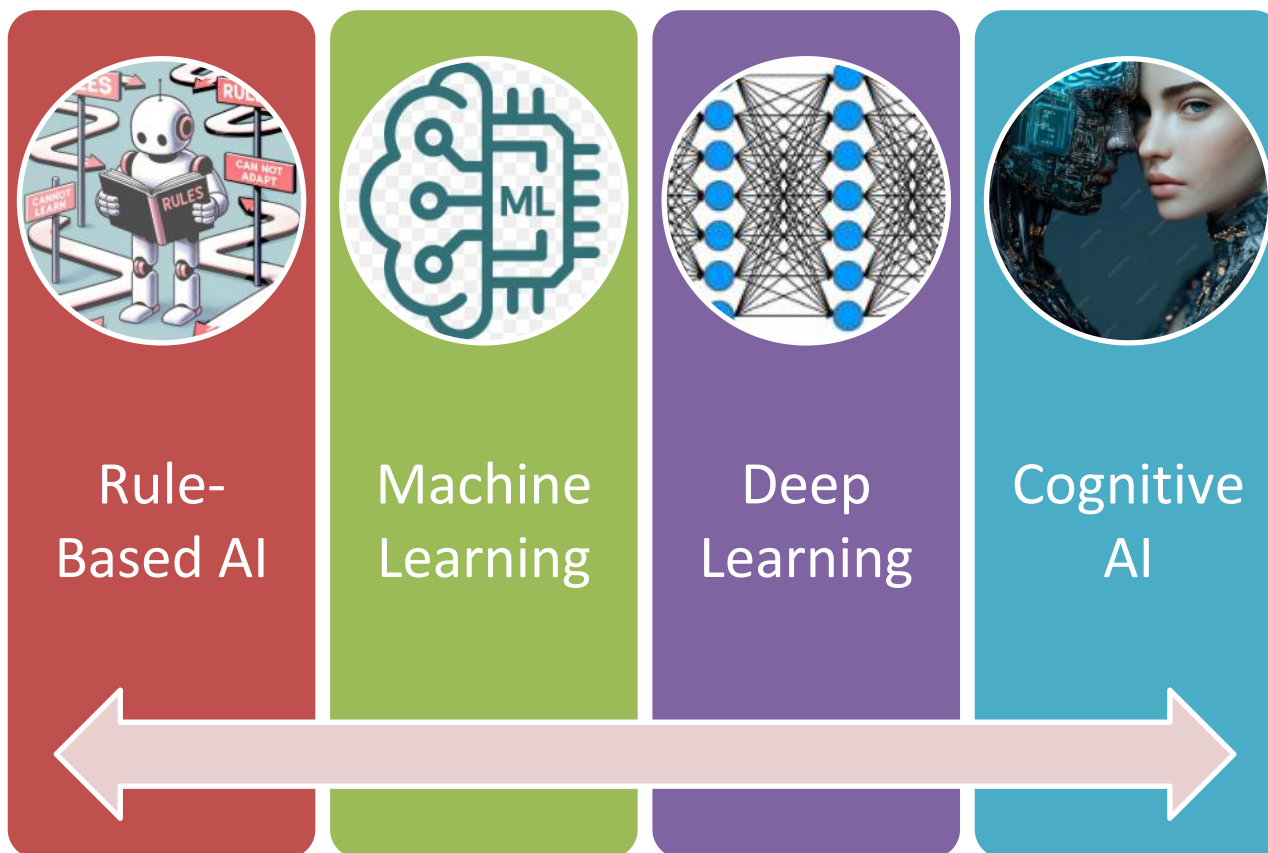
History of AI

- The field of Artificial Intelligence was officially born at the Dartmouth Summer Research Project on Artificial Intelligence (1956), held at Dartmouth College, Hanover, New Hampshire, USA.
- John McCarthy, one of the founders of AI, defined it as: "The science and engineering of making intelligent machines, especially intelligent computer programs."

History of AI

- 1950s: Turing Test
- 1955–56: Logic Theorist
- 1956: Dartmouth Conference
- 1960s-70s: Problem-solving, symbolic AI
- 1980s: Expert systems
- 1990s-2000s: Machine Learning
- 2010s-Present: Deep Learning, Robotics, NLP

Evolution of AI



Evolution of AI

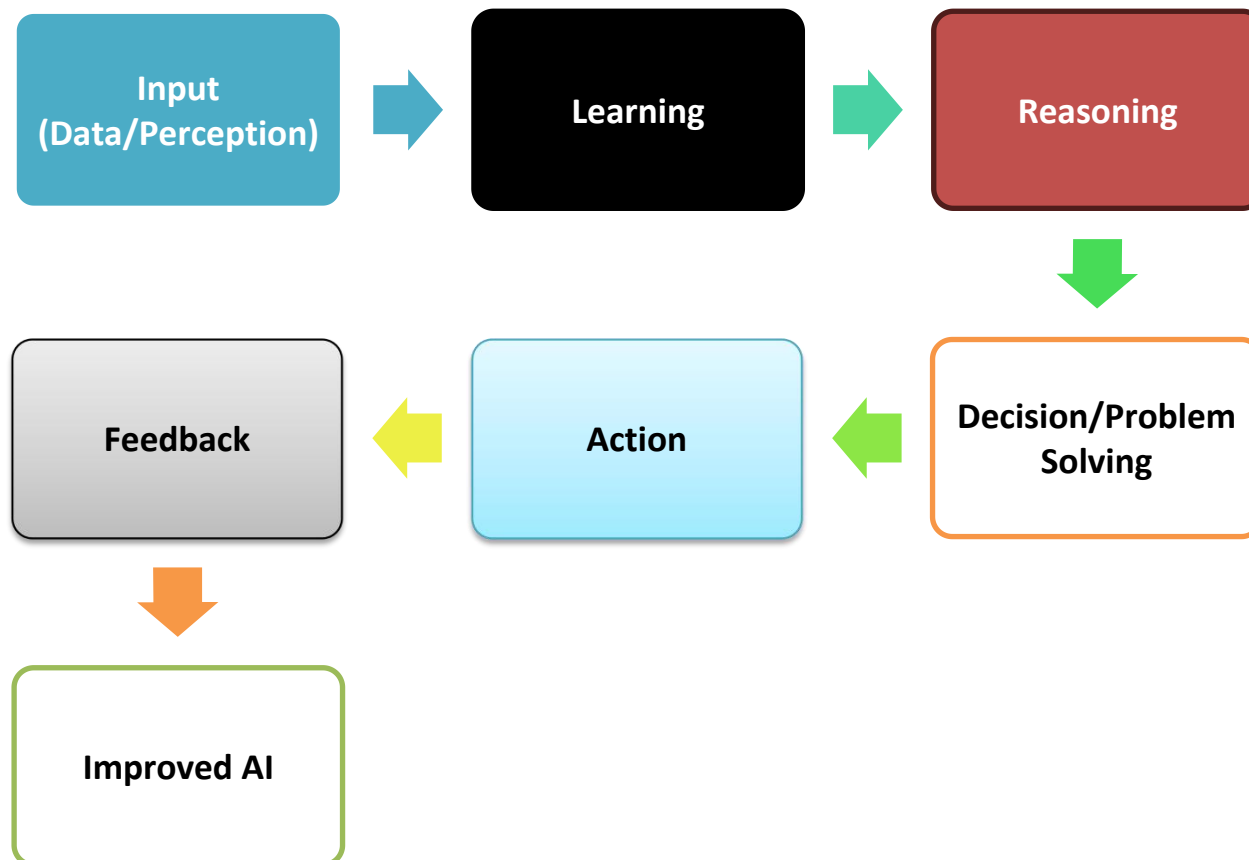
Stage	Period	Key Approach	Description
1. Rule-Based AI	~1950s–1980s	Symbolic AI / Expert Systems	AI built on manually coded rules. If-X-then-Y logic. No learning.
2. Machine Learning (ML)	~1990s–2010s	Data-Driven Learning	AI starts learning from data (e.g., decision trees, support vector machines).
3. Deep Learning (DL)	~2012–Now	Neural Networks	Uses multi-layered neural networks to learn from big data (e.g., image, voice).
4. Cognitive AI	Emerging (Now–Future)	Human-Like Thinking	AI aims to reason , learn , remember , and adapt like humans. Inspired by psychology.

Main Components of an AI System

- **Learning** – Acquiring knowledge from data (e.g., Machine Learning, deep learning, neural networks.)
- **Reasoning** – Drawing conclusions from rules or data
- **Problem Solving** – Finding solutions to complex issues
- **Perception** – Interpreting sensory input (e.g., vision, sound)
- **Action** – Converts decisions into **real-world actions**.
- **Feedback** – AI improves over time by using feedback from the environment.

Main Components of an AI System

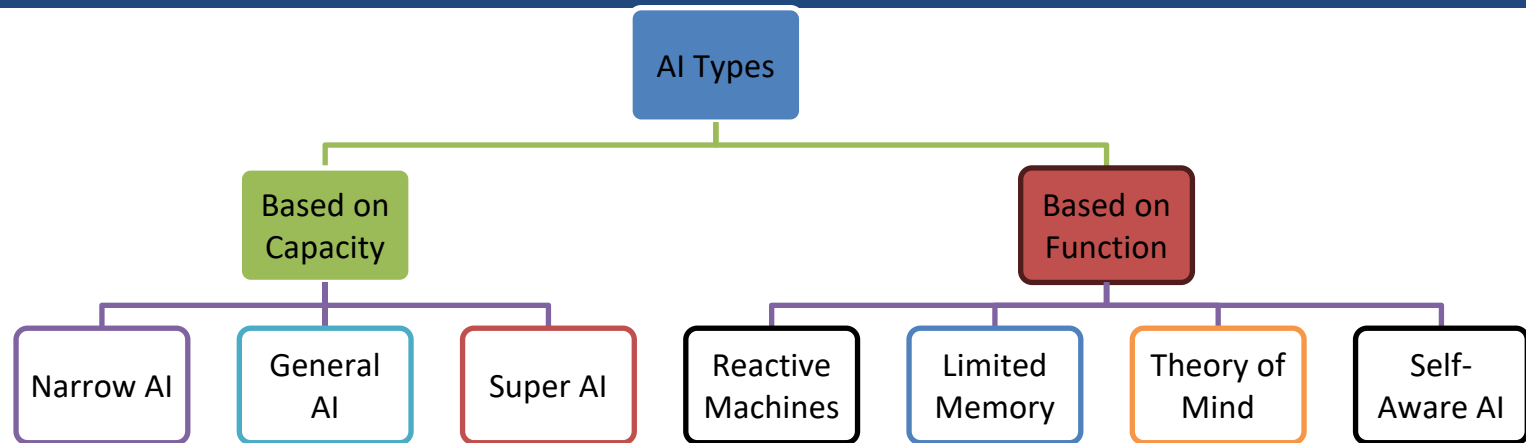
- An **AI Agent/System** usually has these parts:



Main Components of an AI System

- Think of AI like a **student**:
 - **Learning** = studying textbooks (data).
 - **Reasoning** = thinking about problems.
 - **Problem-solving** = answering exam questions.
 - **Perception** = reading the question paper.
 - **Action** = writing the answer.
 - **Feedback** = getting graded and learning from mistakes.

Types of AI



Types of AI Based on Capacity

Type	Capacity	Examples	Status
Narrow AI	Single-task	Siri, Netflix recommendations	Existing
General AI	Multi-task human-level	Hypothetical	Theoretical
Super AI	Super-human	Skynet, HAL 9000	Speculative(theory)

Types of AI Based on Function

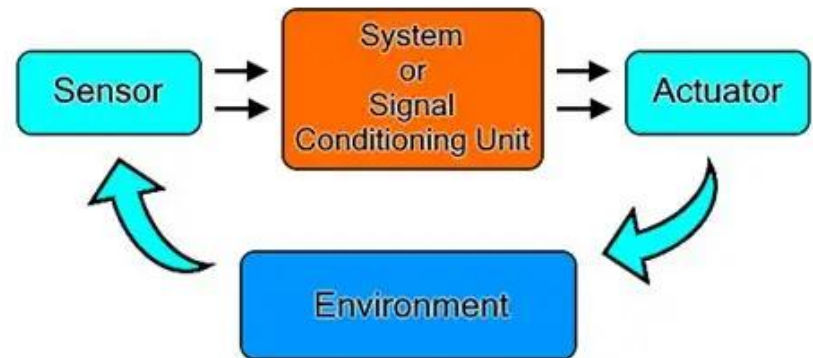
Type	Function	Examples	Status
Reactive Machines	Responds to inputs only	Deep Blue	Existing
Limited Memory	Learns from past experiences	Self-driving cars	Existing
Theory of Mind	Understands human emotions	Future AI companions	Experimental
Self-Aware AI	Conscious AI	Sci-fi AI	Theoretical

AI Agents

- An AI agent is any system that perceives its environment, reasons about it, and takes actions to achieve goals.
- Can be software-only (no physical body).
- May exist in virtual environments, like stock trading bots, recommendation systems, or chatbots.
- Focus is on intelligence and decision-making, not physical interaction.
- Example:
 - ChatGPT (acts on text input to provide answers)
 - Self-driving car's software (decides navigation and safety measures)

Building Block of Intelligent System

- Agent
- Environment
- Sensors
- Actuators
- Effectors
- Goals



Building Block of Intelligent System

- **Agent**

- The central decision-making system.
- It receives input from the sensors and sends output to the actuators.

- **Environment**

- The external world in which the agent operates.
- It changes in response to the agent's actions and other external events.

- **Sensors**

- Devices or modules that perceive information from the environment.
- Example: A camera, microphone, temperature sensor, etc.
- This information is sent to the agent.

- **Effectors**

- Are the agent's interface with the world (e.g., limbs, wheels, speakers).

Building Block of Intelligent System

- **Actuators**
 - Mechanisms the agent uses to take action in the **environment**.
 - Example: Motors, speakers, robotic arms, etc.
- **Goals**
 - Objectives or tasks the agent is designed to achieve.
 - These drive the agent's decision-making process.
- **Real-World Example: Robot Vacuum**
- **Sensors:** Cameras, bump sensors, LIDAR.
- **Agent:** Navigation software.
- **Goals:** Clean the floor efficiently.
- **Actuators:** Motors, speakers.
- **Effectors:** Wheels, vacuum brush, speaker grille.

PEAS Model

- Performance, Environment, Actuators, Sensors
- Example: Self-driving car



PEAS Model

Component	Description	Example (Self-driving car)
Performance measure	What defines success	Safe, fast, fuel-efficient travel
Environment	The surrounding world	Roads, traffic, weather
Actuators	How the agent acts	Steering, brakes, accelerator
Sensors	How the agent perceives	Cameras, radar, GPS

Robot

- A robot is a physical machine capable of carrying out actions in the real world.
 - Usually has actuators (motors) and sensors (camera, lidar, touch).
 - Can operate with or without AI. Some robots follow pre-programmed instructions without learning.
 - Focus is on physical execution of tasks.
- **Example:**
 - Roomba (robot vacuum cleaner)
 - Boston Dynamics' Spot robot
 - Industrial robotic arms in factories

AI Agents vs Robot

Feature	AI Agent	Robot
Physical body	Usually none (software-based)	Has a physical body (hardware)
Environment	Can be virtual or real	Primarily real-world environment
Purpose	Decision-making, reasoning, acting logically	Executing tasks in the real world, may or may not be intelligent
AI dependency	Can be purely AI-based	May use AI for autonomy, but not necessary
Example	Chatbots, AI recommendation system	Industrial robot arm, humanoid robot

AI Agents

- AI agent: perceives environment, acts to achieve goals
- Types:
 1. Simple reflex
 2. Model-based
 3. Goal-based
 4. Utility-based
 5. Learning based
- **Simple Reflex Agent:** Acts based on current perception
- **Model-Based Agent:** Maintains internal state
- **Goal-Based Agent:** Acts to achieve specific goals
- **Utility-Based Agent:** Maximizes a performance measure
- **Learning based:** Improve their performance based on past experiences

Knowledge Representation(KR)

- AI stores and manipulates knowledge to reason and decide
- is how an AI system stores and uses knowledge to reason and make decisions.
- KR Forms:
 - Logic
 - Semantic Networks
 - Frames
 - Ontologies
 - Rule
 - Procedural
- Types of Knowledge
 - Declarative Knowledge – Facts (e.g., "Paris is the capital of France")
 - Procedural Knowledge – How-to (e.g., how to ride a bicycle)
- **Importance of KR**
 - Helps AI systems make decisions
 - Enables understanding and reasoning
 - Forms the backbone of expert systems

Knowledge Representation

- **Logic:** Propositional and predicate logic
- **Semantic Networks:** Nodes represent concepts; edges represent relationships
- **Frames:** Structured representation of stereotypical objects
- **Ontologies:** Formal definitions of concepts and relationships in a domain
- **Rules (If-Then)** – Used in expert systems
- **Procedural** – How-to knowledge

Knowledge Representation

- **Example:**

- Semantic network: Dog → Mammal → Animal
- Frame: Person {Name, Age, Occupation}
- Logic-Based : `If Rain then Wet Ground
- Rules : If rainy then carry umbrella
- Ontologies : Animal > Mammal > Dog
- Procedural : Recipe steps

AI Applications

- Healthcare: Diagnosis, drug discovery
- Transportation: Self-driving cars
- Finance: Fraud detection, trading
- Education: Personalized learning
- Manufacturing: Robotics

Advantage of AI

Advantage	Example
Automation	Factory robots, assembly lines
Accuracy	Tumor detection in medical imaging
24/7 Availability	Chatbots, virtual assistants
Big Data Handling	Weather forecasting, stock predictions
Cost Efficiency	Automated warehouses, predictive maintenance
Learning	Spam filters improving over time
Complex Problem Solving	Self-driving cars, protein folding
Decision Support	AI-assisted diagnostics, financial insights

Limitation of AI

Limitation	Example
Emotions	Chatbots can mimic empathy but don't feel
Common sense	Self-driving car fails in unusual traffic
Creativity	AI-generated art = learned patterns
Ethics	AI cannot morally judge situations
Adaptation	Struggles with unseen scenarios
General intelligence	AlphaGo cannot drive a car
Physical action	AI software alone can't move objects
Context understanding	Misinterprets sarcasm or jokes

Summary

- AI enables intelligent machine behavior
- Types: Narrow, General, Super
- Applications: Healthcare, finance, transport, education, manufacturing
- Agents use PEAS framework
- Knowledge representation is crucial